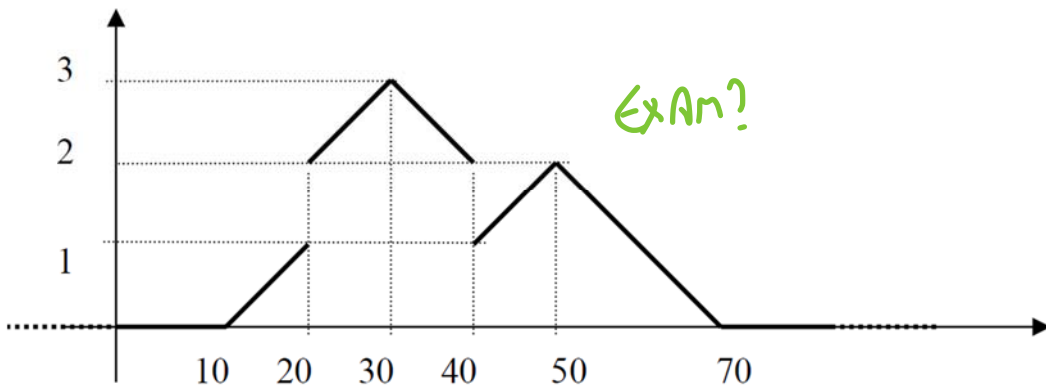


EXERCISES E2

Exercise 1

It is given the signal $s(t)$ represented in the figure below, in which the times (on the x-axis) are expressed in microseconds:



Write the Fourier transform of the $s(t)$ signal.

Calculate the energy of the portion of the signal between 0 and 40 μs .

Without performing the relevant calculation, can you estimate whether the energy of the remaining portion (from 40 μs onward) is greater or less than the energy in the previous tract?

How large is the coefficient a_0 of the Fourier series associated with the periodic signal $s_p(t)$ that is obtained by replicating infinite times the part of $s(t)$ between 0 and 80 μs ?

Exercise 2

Assume that the signal $s(t)$ has a spectral density of energy

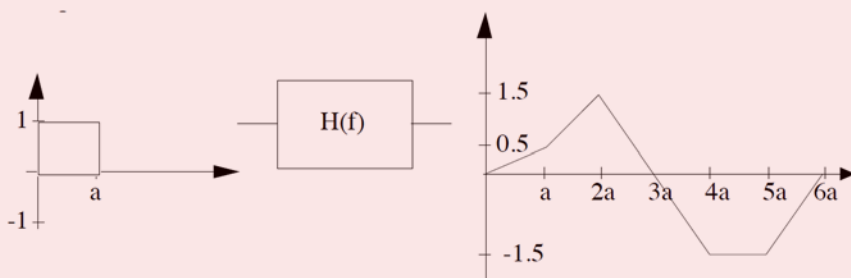
$$G(f) = 4b^2 \text{sinc}^2(fa) \cos^2(2\pi fa)$$

Calculate the autocorrelation of $s(t)$.

Furthermore, knowing that the phase of the transform of $s(t)$ is everywhere zero, exploit this result to derive the original expression of the signal $s(t)$.

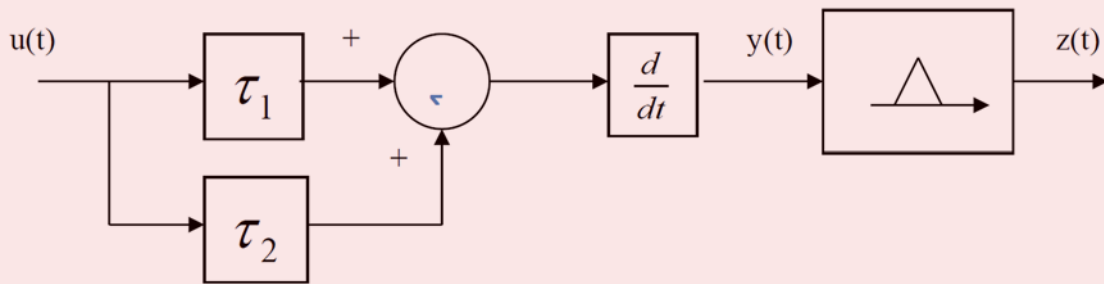
Exercise 3

Consider the system in the figure. First identify its impulse response, and then calculate its the transform to obtain the transfer function.



Exercise 4

The following circuit is given:



The input signal is $u(t) = 1,5 \text{rect}\left(\frac{t-1}{2}\right)$.

The two blocks on the left realize a delay in the signal, of magnitude $\tau_1= 2$ and $\tau_2=3$, respectively.

Write the expression of the spectrum in frequency of $y(t)$.

Finally, knowing that the last block on the right side of the chain has an impulse response of a triangle signal, centered in the time origin, of height 4 and duration 1, determine the value of $z(t)$ at instant $t=2.25$.